

DBMS Study Notes: ON DELETE CASCADE

Comprehensive Guide to Referential Actions, Cascading Deletes & Exam Patterns

1. Purpose & Core Definition

In Relational Database Management Systems (RDBMS), a Foreign Key constraint ensures that data relationships between tables remain valid. However, when a row in the referenced (parent) table is deleted, dependent rows in referencing (child) tables risk becoming **orphan records**.

What is ON DELETE CASCADE? ☐☐☐

ON DELETE CASCADE is an automated referential action associated with a Foreign Key constraint. When a row in the parent (referenced) table is explicitly deleted, the DBMS automatically deletes all corresponding matching rows from the child (referencing) tables.

Primary Objective

Core Objective: Automatically maintain **Referential Integrity** without requiring manual multi-table SQL cleanup scripts.

2. Motivational Example: Employee Schema

Consider a corporate HR database with one primary parent table and two dependent child tables:

Parent Table: Employee

ID (PK)	Name
1	Ram
2	Shyam

Here, *Employee.ID* acts as the Primary Key (PK).

Child Table 1: Salary_Detail

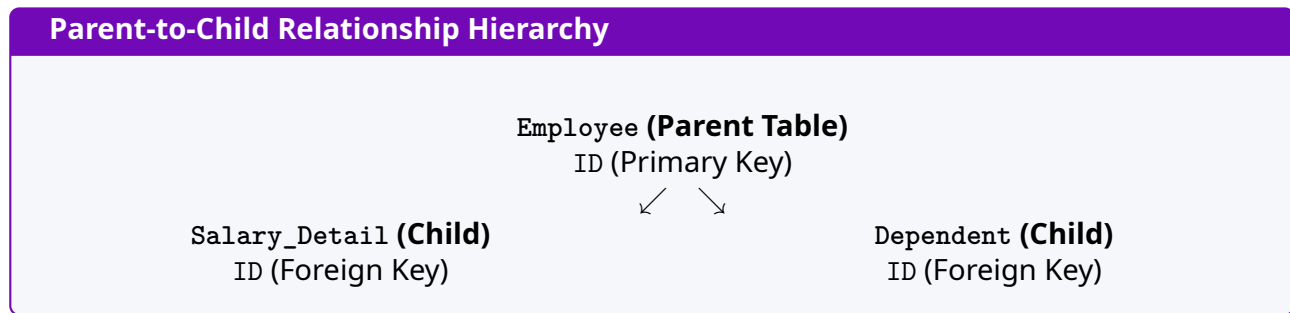
ID (FK)	Salary
1	50000
2	60000

Child Table 2: Dependent

ID (FK)	Dependent_Name
1	Ravi
2	Priya

3. Schema Relationship Architecture

The structural entity topology connects a single parent entity to multiple child records:



4. The Problem Without ON DELETE CASCADE

Suppose employee ID = 1 (Ram) leaves the organization and we execute:

```
DELETE FROM Employee WHERE ID = 1;
```

Default Behavior (RESTRICT / NO ACTION):

1. The DBMS inspects Salary_Detail and Dependent.
2. It detects active rows with ID = 1 referencing the parent.
3. Deleting ID = 1 from Employee would leave orphaned child rows referencing a non-existent employee.
4. **Result:** The DBMS halts execution and throws a **Referential Integrity Constraint Violation Error**.

What is an Orphan Record?

An **orphan record** is a child table row whose Foreign Key points to a Primary Key value that has been deleted from the parent table.

5. State Transformation with ON DELETE CASCADE

When ON DELETE CASCADE is defined on both Foreign Keys, executing:

```
DELETE FROM Employee WHERE ID = 1;
```

triggers an automated multi-table removal state transition:

Before Execution

Employee (Parent)		Salary_Detail (Child 1)		Dependent (Child 2)	
ID	Name	ID	Salary	ID	Dependent_Name
1	Ram	1	50000	1	Ravi
2	Shyam	2	60000	2	Priya

After Execution

Employee		Salary_Detail		Dependent	
ID	Name	ID	Salary	ID	Dependent_Name
2	Shyam	2	60000	2	Priya

Why is it called "CASCADE"?

The term **CASCADE** signifies directional propagation. The deletion of a single root parent row cascades down like a waterfall through all associated child branches.

6 SQL DDL Syntax Implementation

To configure cascading deletions, specify `ON DELETE CASCADE` during table creation or alteration.

6.1 Creating Parent & Child Tables

```
-- Step 1: Create Parent Table
CREATE TABLE Employee (
    ID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL
);

-- Step 2: Create Child Table 1 with CASCADE
CREATE TABLE Salary_Detail (
    ID INT,
    Salary DECIMAL(10,2),
    CONSTRAINT FK_Salary_Employee
        FOREIGN KEY (ID) REFERENCES Employee(ID)
        ON DELETE CASCADE
);

-- Step 3: Create Child Table 2 with CASCADE
CREATE TABLE Dependent (
    ID INT,
    Dependent_Name VARCHAR(50),
    CONSTRAINT FK_Dependent_Employee
        FOREIGN KEY (ID) REFERENCES Employee(ID)
        ON DELETE CASCADE
);
```

6.2 Adding Cascading Action to Existing Table

```
ALTER TABLE Dependent
ADD CONSTRAINT FK_Dependent_Employee
FOREIGN KEY (ID) REFERENCES Employee(ID)
ON DELETE CASCADE;
```

7 Operational Analysis Matrix: Parent vs. Child

7.1 INSERT Validation Examples

Assume `Employee.ID` currently contains {1,2}:

Table 1: Operational Behavior Matrix for DML Operations

Operation	Target Table	Execution Outcome	Referential Rule
INSERT	Parent Table	Always Allowed	Parent exists independently. FK value must exist in Parent (or be NULL).
INSERT	Child Table	Check Parent FK	
DELETE	Child Table	Always Allowed	Removing child rows never invalidates parent.
DELETE	Parent Table	Cascades Child Rows	Child rows automatically removed when ON DELETE CASCADE is active.

• **Valid Insert:**

```
INSERT INTO Salary_Detail VALUES (1, 50000); -- OK (ID = 1 exists in Parent)
```

• **Invalid Insert (Violation):**

```
INSERT INTO Salary_Detail VALUES (10, 50000); -- REJECTED (ID = 10 does not exist)
```

8. Manual Deletion vs. ON DELETE CASCADE

Without CASCADE (Manual Method)	With ON DELETE CASCADE
Requires explicit script sequence: 1. DELETE FROM Salary_Detail WHERE ID = 1; 2. DELETE FROM Dependent WHERE ID = 1; 3. DELETE FROM Employee WHERE ID = 1;	Single parent deletion command: DELETE FROM Employee WHERE ID = 1;
Risk: Missing a child table causes script failure or leaves orphan records if constraints are missing.	Benefit: DBMS automatically handles dependent deletions in transactional integrity.

9. Multi-Level Cascading Chains

ON DELETE CASCADE operates transitively down multi-tier table hierarchies:

Multi-Tier Cascading Architecture

Company (Comp_ID) $\xrightarrow{\text{CASCADE}}$ Department (Dept_ID, Comp_ID) $\xrightarrow{\text{CASCADE}}$ Employee (Emp_ID, Dept_ID)

Transitive Cascade Execution

Executing DELETE FROM Company WHERE Comp_ID = 5; automatically:

1. Deletes linked departments from Department.
2. Deletes all employees belonging to those deleted departments from Employee.

10 10. Target Exam & Interview Q&A

Q1. What is the primary purpose of ON DELETE CASCADE?

To automatically delete related child records when a referenced parent record is deleted, thereby maintaining referential integrity.

Q2. Which integrity constraint is enforced by ON DELETE CASCADE? Referential Integrity.

Q3. With which key constraint is ON DELETE CASCADE associated?

It is defined on a **Foreign Key** constraint in the child (referencing) table.

Q4. Does deleting a record from a child table cause its parent record to be deleted?

No. Cascading works exclusively from ****Parent → Child****, never from ****Child → Parent****.

Q5. Which row is deleted first during a cascading delete?

The DBMS explicitly targets the parent row, and the internal engine automatically purges the corresponding child rows.

Q6. What happens if you try to insert a child row with a non-existent Foreign Key?

The operation is rejected with a **Referential Integrity Violation**, regardless of whether ON DELETE CASCADE is set.

Q7. How does ON DELETE CASCADE differ from ON DELETE SET NULL?

CASCADE completely deletes the dependent child rows, whereas SET NULL retains the child rows but sets their Foreign Key attributes to NULL.

11 11. One-Minute Revision & Exam Cheat Sheet

Ultimate ON DELETE CASCADE Cheat Sheet

DIRECTION $\implies$ Parent Deletion $\longrightarrow$ Automatic Child Deletion
REVERSE FLOW $\implies$ Child Deletion $\longrightarrow$ Parent remains completely unchanged
CONSTRAINT $\implies$ Associated strictly with Foreign Keys (Referential Integrity)

Quick-Check Matrix for Exam Questions:

- Delete Child $\rightarrow$ **ALWAYS SAFE (No Cascade Needed)**
- Insert Parent $\rightarrow$ **ALWAYS SAFE**
- Delete Parent (Default) $\rightarrow$ **REJECTED if child exists**
- Delete Parent (CASCADE) $\rightarrow$ **SUCCESSFUL (Child rows deleted automatically)**